Worked solutions — class questions and practice

Only consult these solutions after completing the relevant classwork. Check and correct your own reasoning. Homework tasks and homework answers are not included.

Multiple-choice (1–5)

1. B

2. C

3. B

4. C

5. B

Graphs and written answers

6. Plot (0,0), (1,3), …, (5,15) and draw a straight line. Equation: gradient = $\Delta x/\Delta t$. Substitute 15 m/5 s. Answer: +3 m/s.

7. Line joining (0 s,+8 m) and (4 s,−4 m). Equation: gradient = $\Delta x/\Delta t$ = [−4−(+8)] m /4 s. Answer: −3 m/s.

8. (a) Horizontal x–t at +4 m; (b) straight upward x–t; (c) x rises, reaches a maximum and then falls, with velocity changing sign.

9. Straight line from (0 s, 8 m/s) to (4 s,0 m/s). Equation: $a=\Delta v/\Delta t$. Substitute (0−8) m/s /4 s. Answer: −2 m/s².

10. Velocity is −2 m/s at every instant, acceleration zero. The position–time graph is a straight line with negative slope.

11. Draw a tangent to the smooth curve at t=3 s; calculate rise/run using two well-separated points on that tangent. The result is instantaneous velocity in m/s.

12. Extension (not assigned homework): signed area = (+4 m/s)(3 s) + (−2 m/s)(2 s) = +8 m net displacement.

13. Flat position–time: stand still. Upward slope: move away from sensor steadily. Downward: move toward steadily. Steepening upward: move away increasingly quickly.